Muhammad Ibrahim

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SUMMARY

M.Sc. candidate in Electrical and Computer Engineering focused on 5G/6G wireless systems, physical-layer modeling, and simulation-based performance analysis. 3+ years of research experience building Python, MATLAB, and C/C++ workflows for MIMO, beamforming, channel modeling, and multi-user radio systems. Co-author of 4 research publications in IEEE. Interested in baseband system development, link-level and system-level simulation, RRM, and future radio access technologies.

EDUCATION

University of Manitoba — 2024 to July 2026 Expected
M.Sc. in Electrical and Computer Engineering, ***CGPA: 4.38/4.5*** — Winnipeg, Canada

Relevant coursework: Advanced Wireless Communication, Signal Compression and Coding, Estimation Theory, Deep Generative Modeling, Reinforcement Learning (Audit)

National University of Sciences and Technology — 2020 to 2024
B.Sc. in Electrical Engineering, ***CGPA: 3.66/4.0*** — Islamabad, Pakistan

Relevant coursework: Wireless Communication, Digital Signal Processing, Linear Control Systems, Data Structures and Algorithms, Embedded Systems

RESEARCH EXPERIENCE

Wireless Communication, Networking and Services Lab — Sep. 2024 to Present
Graduate Research Assistant, University of Manitoba — Winnipeg, Canada

- **Project 1:** Developed the mathematical system model for stacked intelligent metasurface-assisted downlink transmission, including multi-user channels, shared SIM phase shifts, transmit beamforming, inter-user interference, and achievable-rate formulation.
- Implemented deep unfolding methods by mapping iterative optimization updates into trainable layers, then evaluated convergence speed, sum-rate performance, robustness, and computational cost against conventional optimization baselines.
- **Project 2:** Developed DRL-based methods for communication graph selection and consensus behavior in multi-agent systems, with a focus on reducing communication cost while maintaining stable convergence.
- Tested graph-based control policies under changing network topologies, local communication constraints, and safety-aware reward terms using convergence error, communication overhead, and policy stability as evaluation metrics.
- Built and maintained simulation code in Python, MATLAB, PyTorch, C/C++, and Linux for wireless system modeling, Monte Carlo evaluation, plotting, debugging, and IEEE publication results.
- Worked with 3GPP-inspired wireless channel models, including path loss, fading, antenna effects, beamforming, Doppler, and SINR-based performance evaluation.

Information Processing and Transmission Lab — Jun. 2023 to May 2024
Undergraduate Research Assistant, NUST — Islamabad, Pakistan

- Built Python simulation scripts for wireless resource allocation and intelligent reflecting surface-assisted communication networks.
- Used GDAL-based 3D environment data to compare network behavior across urban, suburban, and rural deployment settings.
- Tested DDQN-based radio parameter selection for dynamic multi-user scenarios and compared results with baseline approaches.

ENGINEERING EXPERIENCE

Team Burraq — 2022 to 2024
AI and Electrical Systems Member, Student-Led UAV Competition Team — Islamabad, Pakistan

- Worked on autonomous UAV payload delivery systems as part of a student engineering competition team.
- Integrated sensors, cameras, and embedded hardware interfaces including UART, I2C, GPIO, and ADC.
- Used telemetry logs, sensor readings, and field-test observations to debug hardware and software issues.

SELECTED PROJECTS

5G/6G Wireless System Simulation Python, MATLAB, C/C++

- Built simulations for multi-user MIMO, beamforming, intelligent surfaces, and radio performance analysis.
- Implemented channel generation, optimization routines, Monte Carlo runs, and plotting scripts.
- Compared performance across transmit power, antenna settings, user layouts, and channel conditions.

Baseband-Oriented Beamforming Optimization Python, PyTorch, MATLAB

- Implemented deep unfolding models for beamforming and phase-shift optimization in multi-user MISO systems.
- Mapped iterative optimization updates into trainable layers and benchmarked against standard baselines.
- Measured achievable rate, convergence speed, robustness, and computational cost.

DRL-Based Object Detection and UAV Tracking Python, ROS, MAVLink | GitHub

- Built a UAV tracking framework using ROS object detections to guide drone motion toward a target.
- Used image-center error and detected object center error as the control signal for RL-based tracking.
- Implemented DDPG, TD3, and prioritized replay DDPG with custom environment and reward design.
- Connected RL control outputs with MAVLink-based drone movement commands.

Cooperative Safe MADRL Python, PyTorch | GitHub

- Developed a cooperative MADRL framework for UAV trajectory optimization with safety constraints.
- Used Lyapunov-inspired safety terms to reduce inter-UAV collisions during trajectory learning.
- Benchmarked DSMADRL against IPPO and MAPPO using rewards, violations, QoS, and energy metrics.

TECHNICAL SKILLS

Programming and Development: Python, C/C++, MATLAB, Linux, Bash, Git, PyTorch, NumPy, Pandas, Scikit-learn

Wireless and Baseband Systems: 5G/6G, Physical Layer, RRM, OFDM, MIMO, Massive MIMO, Beamforming, Channel Modeling, Interference Analysis, Resource Allocation, 3GPP models

Simulation and Verification: Link-Level Simulation, System-Level Simulation, Monte Carlo Simulation, Performance Analysis, Algorithm Validation, Numerical Optimization, Test Result Analysis

Tools and Hardware Exposure: ns-3, ROS 2, Gazebo, Git, Linux, UART, I2C, SPI, GPIO, ADC, ESP32, Arduino, Jetson Nano

SELECTED PUBLICATIONS

On the Performance of Multi-IRS-Assisted Networks Across Real Urban, Suburban, and Rural Environments
M. Ibrahim, W. H. Raza, M. Moiz, S. A. Hassan, H. Jung, and M. Gidlund.
IEEE WCNC | Paper link

Deep Unfolding for SIM-Assisted Multiband MU-MISO Downlink Systems
M. Ibrahim, A. Mezghani, and E. Hossain.
IEEE Communications Letters | Paper link

Are Stacked Intelligent Metasurfaces Better than Single-Layer Reconfigurable Intelligent Surfaces for Wideband Multi-User MIMO Communication Systems?
M. Ibrahim, A. Mezghani, and E. Hossain.
IEEE Transactions on Wireless Communications | Paper link

Science-Informed Design of Deep Learning With Applications to Wireless Systems: A Tutorial
A. Termehchi, E. Hossain, A. Vera-Rivera, M. Ibrahim, and I. Woungang.
IEEE Communications Surveys and Tutorials | In Review | Paper link

HONORS AND AWARDS

University of Manitoba Graduate Fellowship | International Graduate Student Entrance Scholarship | Research Completion Award | NUST Academic Merit Scholarship, 2021 and 2023 | NUST High Achiever Award, Silver Medal | Winner, UAS Challenge Pakistan 2022